DO NOW — RECALL

Lesson 099 · CISC and RISC: Instruction Set Design

Name: _____ Class: _____ Date: _____

1. What is ISA?

2. What does an opcode identify?

3. What is a load/store instruction?

4. Simplify (A∧B)∨(A∧¬B).

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Instruction Set Architecture—programmer-visible instruction operations.

2. Model answer

Which operation the CPU performs.

3. Model answer

Transfers values between registers and memory.

4. Model answer

$A \wedge (B \vee \neg B) = A$.

By the end, you can…

• CISC typically offers a larger/more complex instruction repertoire, with instructions able to combine operations; RISC generally uses more regular, often simple operations.

• RISC load/store designs explicitly move operands between memory and registers and perform ALU work using registers.

• Avoid outdated absolutes: modern CISC and RISC microarchitectures can both pipeline, use caches and achieve strong performance; trade-offs depend on implementation.

Vocabulary — know these terms

ISA — Programmer-visible machine instruction set.

CISC — Complex Instruction Set Computer.

RISC — Reduced Instruction Set Computer.

load/store — Explicit memory-register data transfer.

NEW LEARNING

Read, annotate and explain

Observe

CISC typically offers a larger/more complex instruction repertoire, with instructions able to combine operations; RISC generally uses more regular, often simple operations.

Apply

RISC load/store designs explicitly move operands between memory and registers and perform ALU work using registers.

Evaluate

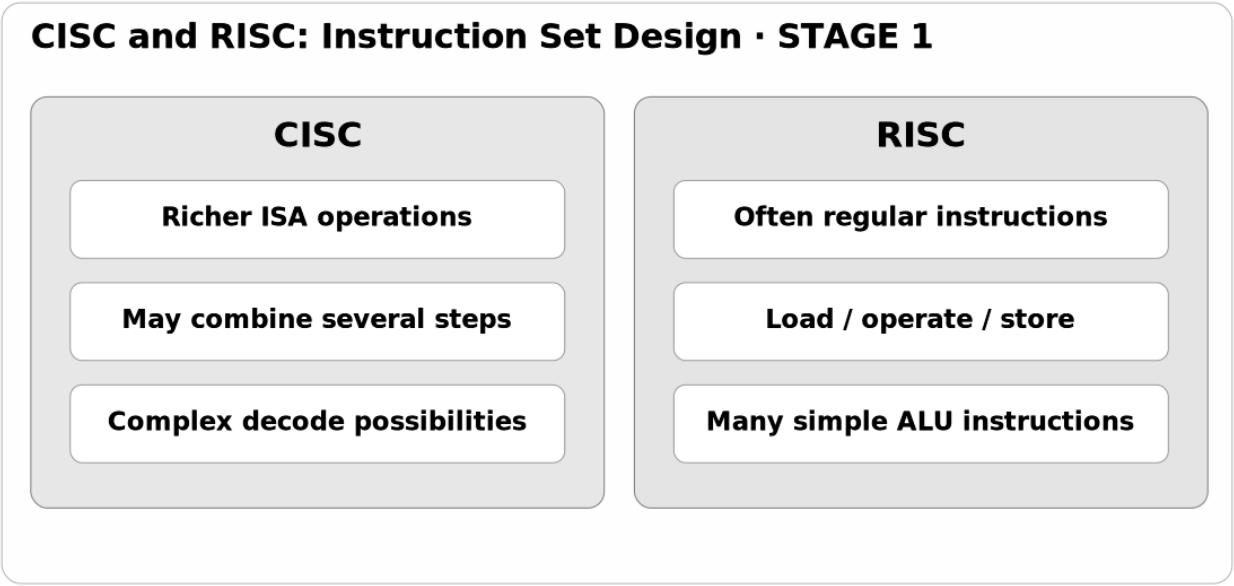
Avoid outdated absolutes: modern CISC and RISC microarchitectures can both pipeline, use caches and achieve strong performance; trade-offs depend on implementation.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

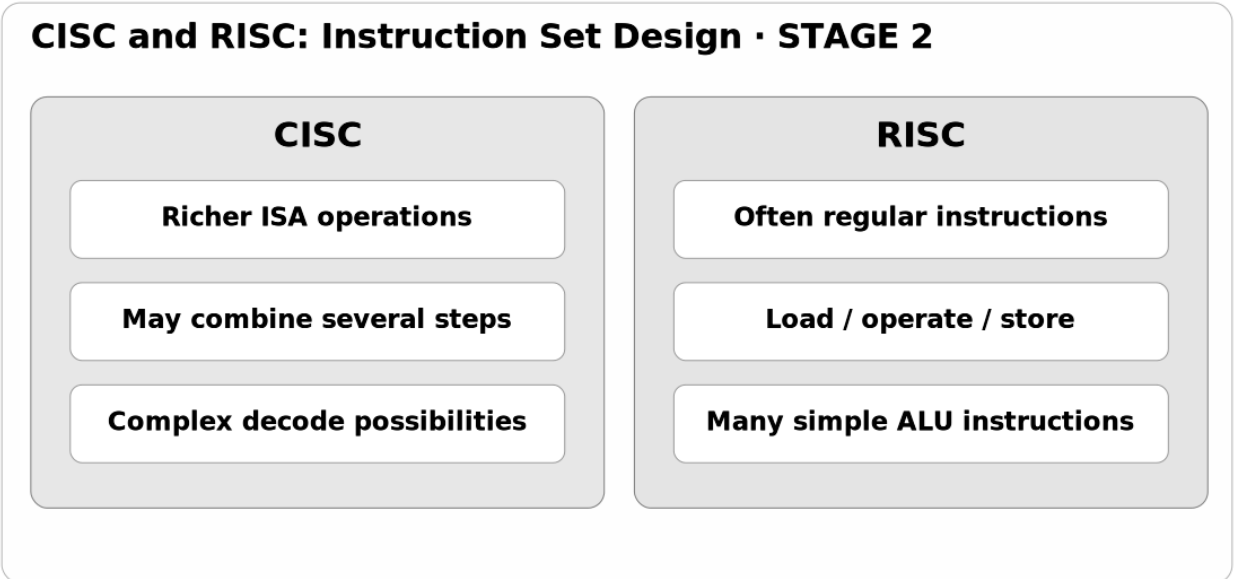
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



CISC and RISC: Instruction Set Design · STAGE 1; CISC: generally richer/variable instruction complexity

New Learning

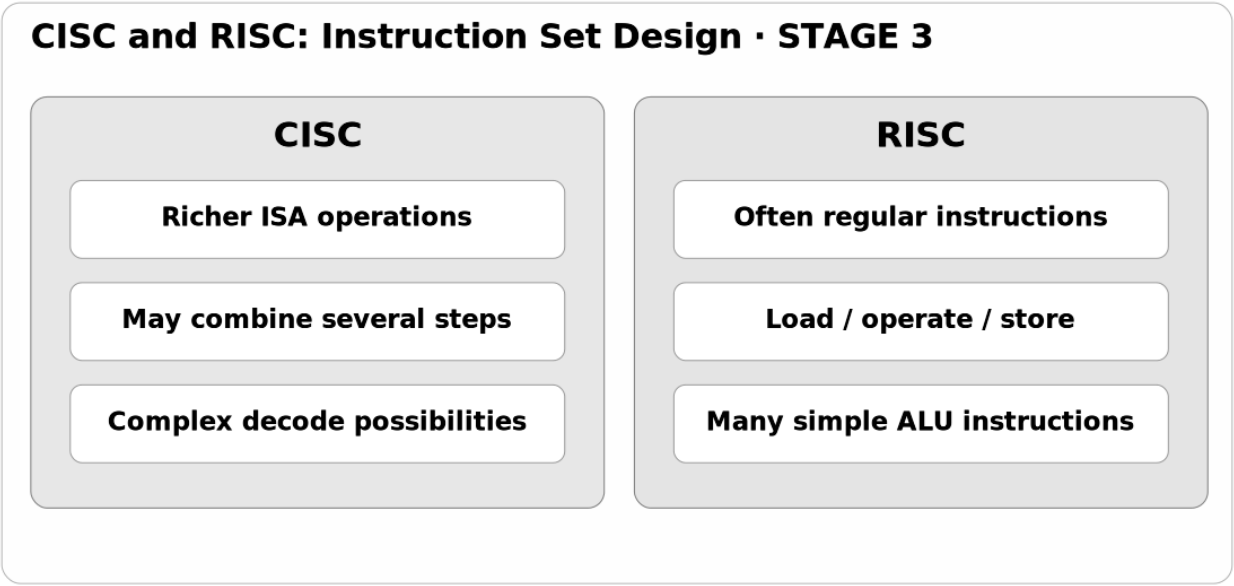


CISC and RISC: Instruction Set Design · STAGE 2; CISC: generally richer/variable instruction complexity; RISC: smaller, regular operations; load/store pattern

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning

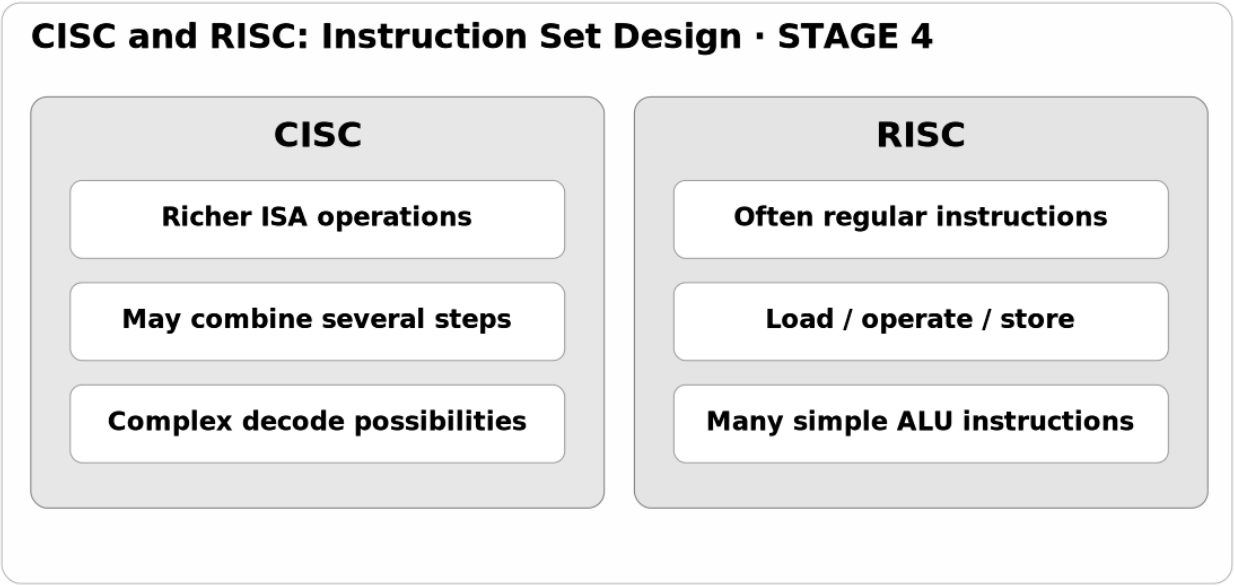


CISC and RISC: Instruction Set Design · STAGE 3; CISC: generally richer/variable instruction complexity; RISC: smaller, regular operations; load/store pattern; Both translate instructions to micro-operations in many modern designs

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



CISC and RISC: Instruction Set Design · STAGE 4; CISC: generally richer/variable instruction complexity; RISC: smaller, regular operations; load/store pattern; Both translate instructions to micro-operations in many modern designs; Compare code size, decoding, power and workload

I DO — WORKED EXAMPLE

Follow the worked method

Model a RISC load-add-store sequence for A5,B8 and result C.

Follow the reasoning

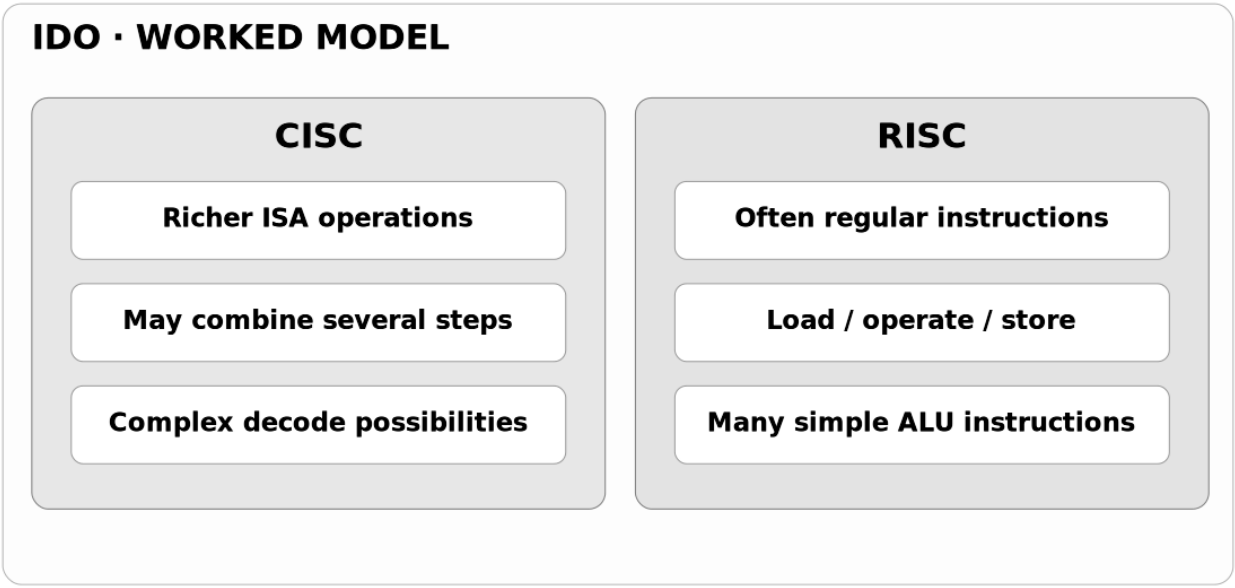
LOAD R1,A→5; LOAD R2,B→8; ADD R3,R1,R2→13; STORE C,R3→13.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Model a RISC load-add-store sequence for A5,B8 and result C.

LOAD R1,A→5; LOAD R2,B→8; ADD R3,R1,R2→13; STORE C,R3→13.



IDO · WORKED MODEL; Model a RISC load-add-store sequence for A5,B8 and result C.; LOAD R1,A→5; LOAD R2,B→8; ADD R3,R1,R2→13; STORE C,R3→13.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

Guided 1 — TRY

For A4,B7, trace the same abstract load/store sequence and state result.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

Question 1

State two common RISC and two common CISC characteristics. [4]

Question 2

Trace the abstract RISC sequence above when A=6,B=9, give R1,R2,R3,C. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

Question 3

Evaluate statement 'CISC processors cannot use pipelining and RISC always uses fewer instructions'. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking.

Modern CISC implementations cannot use pipeline stages.

Circle: Confident / Getting there / Need help

Explain what you can now do independently.

GREEN PEN — Correct and explain

Choose one error. Write the correct method and explain precisely why your first response was wrong.

WE DO — CHECK AND IMPROVE

Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

Guided 1 — CHECK

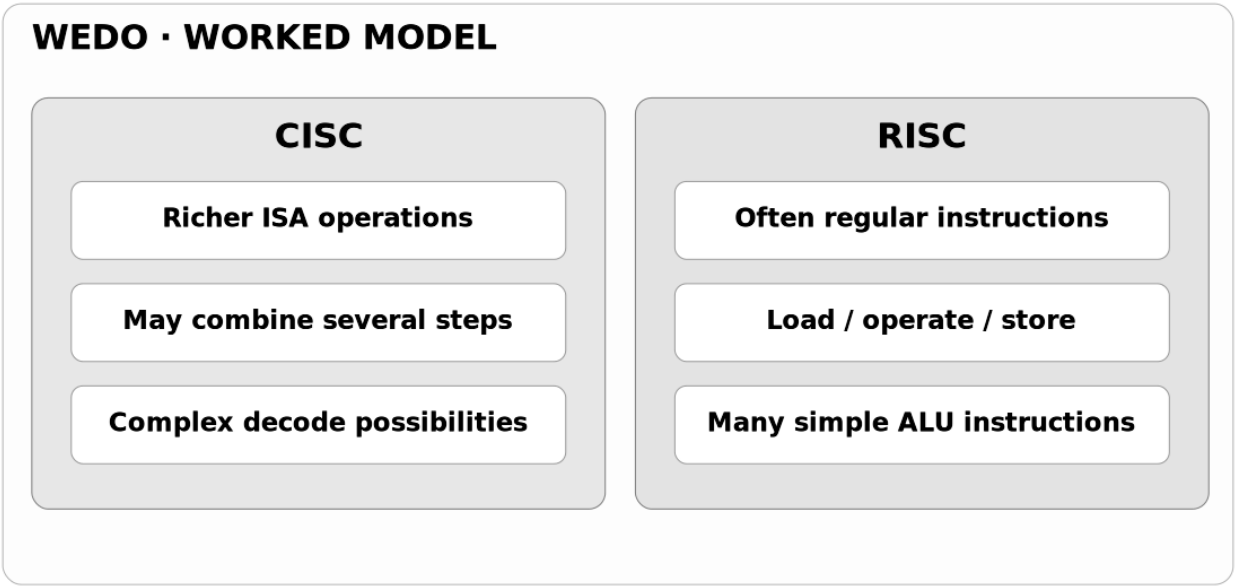
Follow the reasoning
R1=4,R2=7,R3=11, memory[C]=11.

We Do source CHECK · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

For A4,B7, trace the same abstract load/store sequence and state result.

R1=4,R2=7,R3=11, memory[C]=11.



WEDO · WORKED MODEL; For A4,B7, trace the same abstract load/store sequence and state result.; R1=4,R2=7,R3=11, memory[C]=11.